Free balancing equations worksheet worksheet 23

Maths | 8th grade | Algebra

1. If 3(x + 4) = 2(x + 10), what is the value of x?

A) 0
B) 2
C) 4
D) 6

2. What is the value of y if y/3 - 5 = 1?

A) 12
B) 15
C) 18
D) 21

3. The sum of a number and its reciprocal is 5/2. What is the number?

A) 1
B) 2
C) 3/2
D) 5/2

4. Solve for z: 2z - 7 = 5 - z.

A) 1
B) 2
C) 3
D) 4

5. If f(x) = 3x + 2, find f(2) + f(-2).

A) 0
B) 2
C) 4
D) 6

6. What is the value of x if 4(x - 3) = 2x + 10?

A) 8
B) 9
C) 10
D) 11

7. Simplify the expression: 2(3x + 4) - 5x.

A) x + 8
B) x + 4
C) x - 4
D) x - 8

8. If 5p = 35, what is the value of 2p + 7?

A) 14
B) 21

C) 28
D) 35

9. Which of the following is a solution to the equation x^2 - 9x + 20 = 0?

A) 4
B) 5
C) 6
D) 7

10. If h(t) = -2t^2 + 4t + 1, find the value of h(1).

A) 1
B) 3
C) 5
D) 7

Answer Key

1. C
Distribute both sides to get 3x + 12 = 2x + 20, then solve for x by subtracting 2x and 12 from both sides to find x = 8.

2. B
Add 5 to both sides to get y/3 = 6, then multiply by 3 to find y = 18.

3. C
Let the number be x. Then x + 1/x = 5/2. Multiply through by x to find x^2 - 5x + 2 = 0. Solving gives x = 3/2.

4. C
Add z to both sides to get 3z - 7 = 5, then add 7 and divide by 3 to solve z = 4.

5. D
Calculate f(2) = 8 and f(-2) = -4, then add the results to find 8 + (-4) = 6.

6. A
Distribute to get 4x - 12 = 2x + 10, subtract 2x from both sides, then add 12 to solve for x = 11.

7. A
Distribute to get 6x + 8 - 5x, then combine like terms to obtain x + 8.

8. B
Solve for p by dividing by 5 to get p = 7, then substitute into 2p + 7 to find 2(7) + 7 = 21.

9. B
Factor the equation to (x - 4)(x - 5) = 0. The solutions are x = 4 and x = 5.

10. B
Substitute t = 1 to get h(1) = -2(1)^2 + 4(1) + 1 = 3.